Using Machine Learning to Identify Brain Tumours

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Introduction

The most common types of brain tumour are meningiomas, gliomas, and pituitary tumours [1][3]. Gliomas are the least likely of the three to be benign, with 75% of malignant tumours being classed as gliomas [2]. Thus, detecting which type of tumour is present is important for treatment as it can help to determine the likelihood of the tumour being malignant without any invasive measurements [3]. This is especially important as early detection of cancerous tumours can greatly increase the patients chance of survival [4].

Many machine learning and AI based models for detecting these types of tumour have been developed over recent years, all with high levels of accuracy but varying computational costs [1][3][5]. This paper will investigate how convolution neural networks (CNNs) and residual networks (ResNets) perform at identifying these tumour types.

Model Selection

A CNN architecture, which can be seen in Figure 1 was used to classify the brain scans as this was designed for image classification [6]. ResNets follow a similar structure but instead use skipped connections to add a correction to the model at each residual layer. This was also developed for this problem, using the architecture shown in Figure 2, with the aim of validating that a CNN is a better choice of model for this problem, as there is not enough data for deep learning to have a large benefit. Other possible architectures such as multilayer perceptrons and recurrent neural networks were deemed unsuitable due to the need for image classification and lack of sequential data [6]. A mini-batch gradient descent approach was used rather than stochastic gradient descent to reduce the computation time of training the networks.

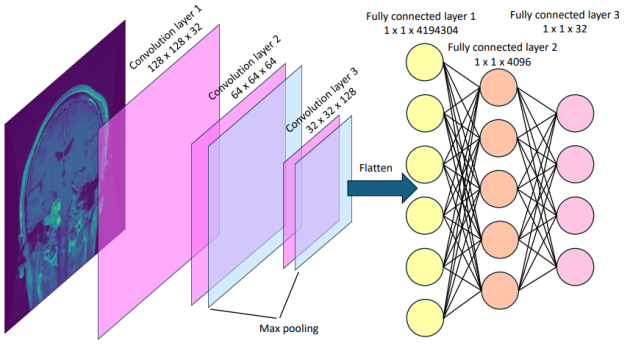


FIG. 1: A diagram of the network architecture of the CNN used.

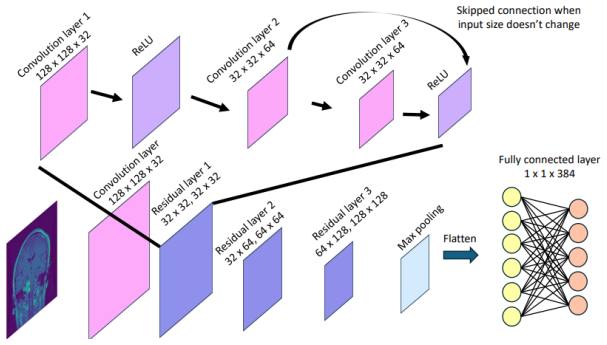


FIG. 2: A diagram of the architecture of the ResNet used.

A max pooling aggregate function was used in the CNN and ResNet. This was chosen over average pooling as it produced higher results across all measured performance metrics. This makes sense as the difference between tumour types can be subtle and specific details need to be conserved in order to construct an accurate model. Adaptive max pooling from the PyTorch library was also tested, however, the lack of specification of stride and kernel size reduced the accuracy of all of the models tested.

The sigmoid activation function was initially used as it is non-linear, and produces values between 0 and 1, which is appropriate for a classification problem such as this, which produces probabilities. This stopped the models converging entirely, and so was replaced with the rectified linear activation function. This did not present the same problem as the upper limit of the gradients are not capped at 1 and its gradient is always either 0 or 1 which helps to stop the vanishing gradient problem in back propagation. In order to normalise the probabilities of belonging to each class for each image, the softmax function was used on the outputs of each model.

A drop out function was also introduced into both architectures to further reduce over-fitting by forcing the model to learn more general characteristics of the tumour types so that when training, the models were not seeing the same images, or augmented versions of the same images every epoch [9]. The drop out rate for each neural network was initially set to 0.05 and was increased until the model was no longer over-fitting when trained with augmented data.

Training The Model

The data set was split $70-15-15$ between training, validation and test data sets. The same data sets were used for every model in order to make the performance metrics more

comparable. Also, having the same amount of each tumour type in the training data set for every model made it easier to notice biases resulting from uneven class splits in the data.

The cross entropy loss function from PyTorch was used to calculate the loss of the training and validation data sets after each epoch. This was chosen as it considers the certainty of each prediction and penalises uncertain predictions, which is useful for this classifier problem.

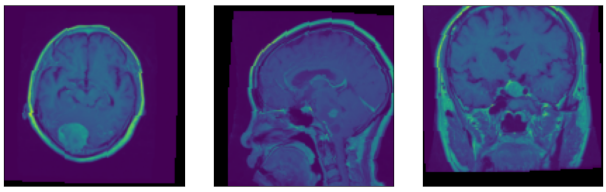
The Adam optimiser was used as it includes learning rate adjustments for each parameter and performs well for problems with high dimensional parameter spaces, such as this [8]. This was used in conjunction with a learning rate scheduler, as with Adam alone, each model either plateaued at a relatively high loss (around 0.7) quickly or it would reach lower validation losses that fluctuated dramatically, rather than producing a low validation loss value and gradually reducing it. The multiplicative learning rate scheduler from the PyTorch library was used so the learning rate started high to produce lower validation losses, and then reduced to gradually reduce the validation loss value further.

The initial learning rate was chosen as the highest learning rate which didn't fluctuate to the point of triggering early stopping within the first 10 epochs so that it would reach as low validation loss. The function used for the multiplicative learning rate was chosen by examining the variation of train and validation loss for many different constant and scheduled learning rates. Repeated fluctuations of the validation were seen as indicators of an excessively high learning rate, this was then used to alter the function so that it reduced to a lower learning rate at this epoch. This was trialled and refined repeatedly until the validation loss curve followed a near smooth descent without plateauing.

To reduce the impact of over-fitting, early stopping was used so the model stops training once the validation loss starts to plateau. The state of the model when the lowest validation loss was produced is restored so that the over-fitting that occurs before the network notices the plateau does not affect the final results.

Data augmentation was introduced to reduce over-fitting so the networks would see slightly different training images at every epoch. This forced the networks to focus on the identifying features of the tumours, which were not altered through augmentation, rather than effectively learning the labels of each training image. The augmentation transformations were chosen as translations and minor rotations to leave the images as realistic tumour scans. The range of the rotations and translations applied was chosen by running the code with a range of parameters and using those which minimised over-fitting. The degree of over-fitting was determined by examining the loss function plots as well as the training and validation data, as shown in Figure 4a. Figure 3 shows examples of images after augmentation. Both architectures were trained with the original training data set and with augmented training data to validate that this decision improved accuracy.

The amount of epochs was chosen as the average epoch when the validation loss plateaued to limit over-fitting. The amount of consecutive epochs where the validation loss did not improve was used to select the patience of each model; the smallest consecutive amount which did not occur during any minor plateaus earlier in training was chosen.



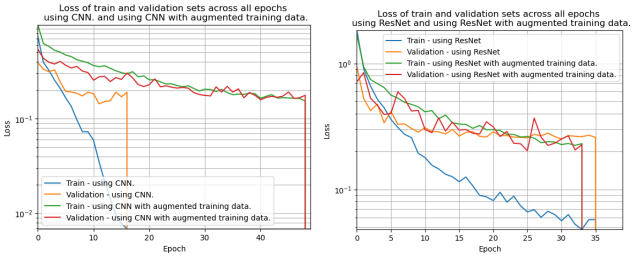
(a) Meningioma tumour (b) Glioma tumour (c) Pituitary tumour

FIG. 3: An example of the transformations applied to the training data set to create the augmented data set.

Model Performance

The true positive rate (TPR) and false positive rate (FPR) of each model was calculated as it considers all three classes at once. A confusion matrix was produced for each model, which shows the most common false positives. For a more detailed analysis, the receiver operating characteristic (ROC) curve of each model for meningioma, glioma, and pituitary tumours, and the areas under these curves (AUCs), were plotted and calculated. This showed the TPR and FPR across many thresholds for each class, providing more detail than the previously calculated TPR and FPR.[6].

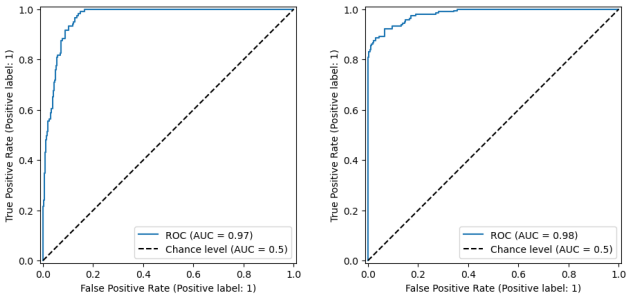
From Figure 4, it can be seen that the data augmentation successfully reduced the models' over-fitting, and, for both CNN and ResNet architectures, this allowed the validation loss to reach lower values. Additionally, these plots also show that the ResNet architecture produced training and validation losses which fluctuated more than the CNN. This is likely due to the batch size used in the model causing excess noise in the batch normalisation process [10].



(a) Convolutional Neural Network (b) ResNet

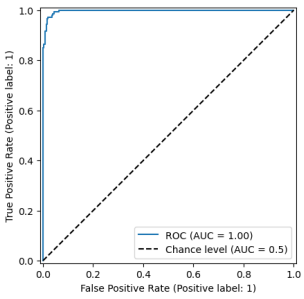
FIG. 4: Comparison of the validation and training losses against epoch when the augmented and original images are used for training.

The ResNet and CNN architectures trained on augmented data performed similarly, as shown in Table I. The confusion matrices, shown in Figure 6, also suggest that the ResNet



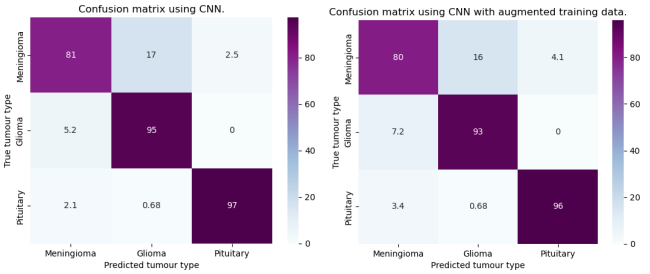
(a) Meningioma

(b) Glioma



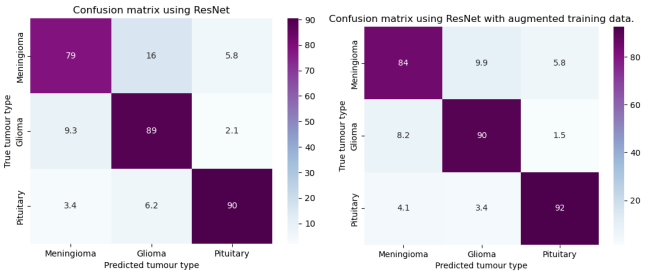
(c) Pituitary

FIG. 5: The ROC curves produced by the CNN trained with augmented data.



(a) Confusion matrix showing the results of training with the original data set with a CNN.

(b) Confusion matrix showing the results of training with the augmented data set with a CNN.



(c) Confusion matrix showing the results of training with the original data set with a ResNet.

(d) Confusion matrix showing the results of training with the augmented data set with a ResNet.

FIG. 6: Comparison of confusion matrices

models were worse at identifying pituitary tumours than the CNN models, but more effective at identifying meningioma and glioma tumours, when trained with augmented data.

Architecture	CNN		ResNet	
Training data	Original	Augmented	Original	Augmented
TPR	0.920	0.905	0.866	0.895
Meningioma AUC	0.977	0.967	0.939	0.963
Glioma AUC	0.985	0.982	0.957	0.978
Pituitary AUC	0.999	0.997	0.989	0.988

TABLE I: The true positive rates and the area under ROC curve for each type of tumour for each model.

The AUCs displayed in Table I and Figure 6 show that every model performed the best when identifying pituitary tumours and the worst at identifying meningioma tumours. As pituitary tumours are characterised by their location in small pituitary gland, it makes sense that they would be the easiest to identify from the training data [11]. The meningioma tumours likely had the least accuracy due to the smaller amount of images in the dataset when compared with glioma and pituitary images.

The ROC curves used to calculate the AUCs for the CNN trained with augmented data can be seen in Figure **??**.

The ResNet trained with augmented data had worse TPR than the CNN, regardless of whether augmented data was used. This suggests that the CNN correctly predicted the categories of more images in the test data set, however, the ResNet makes more accurate predictions when considering a range of thresholds, as it improves on the CNNs AUC for meningioma and glioma tumours.

Contrary to expectations, the CNN without augmented data performed the best by every performance metric displayed in Table I.

Discussion

The performance metrics of all four models agree with previous literature for CNN and shallow ResNet architectures, with TPR around 0.85 - 0.95 and AUCs > 0.9 [1]. The ResNet produced a lower TPR and AUC for all tumour types, as expected, and took over twice the training time compared to the CNN, which was not expected from previous studies [1]. This is likely due to the lack of depth in this network as it consists of 15 layers. Additionally the data set used is likely too small to reap the benefits of ResNet over CNN, and the vanishing gradient problem is controlled by the rectified linear activation function, causing CNNs to be as, if not more effective. Both CNN models performed better than the ResNet models, which validates the choice of the CNN model. [7]

When trained with the original non-augmented training data set, the ResNet performed worse on every performance metric shown in Table I. Figure 4 shows that its validation loss plateaus higher than all other models. As the ResNet is a deeper model, it begins over-fitting, and hence the validation loss begins plateauing, earlier than it does for the CNN.

In contrast, Figure 4 show the data augmentation successfully stopped both models over-fitting and led to much higher TPRs and AUCs for the ResNet but not the CNN. However, the method of choosing augmentation parameters does not guarantee that the full tumour is still included in all of the images. Instead, more images from the augmented data set could be displayed so there is a greater chance of noticing any poor augmentations, however this would be a very time consuming process due to the amount of images. A better approach may be to use a data set which contains the coordinates of the tumour on each image so that the size of possible translations and rotations could be limited to ensure no tumours would be altered by the augmentation.

Figure 4b also shows the validation loss, when not trained on augmented data, plateaued long before early stopping was triggered. This is because small decreases in the validation loss can be seen occasionally throughout the plateau. This suggests that early stopping should be improved by introducing a higher minimum difference between loss values to be considered as decreasing rather than plateauing.

In addition, all four models had the lowest AUC for meningiomas and the highest for pituitary tumours. This could be a result of the training set having an uneven split of tumour types, as all the models had the same amount of each tumour type in the training set. For future models, the augmented data could be used to give the models an equal amount of data of each tumour by augmenting the same images multiple times in different ways.

Differences in the performances of the CNN and ResNet architectures could also be due to one's hyperparameters being optimised less effectively than the other. To mitigate this affecting conclusions, in the future one automated hyperparameter optimisation method, such as Bayesian optimisation, should be used on all architectures to ensure they are all similarly optimised. [12]

As can be seen from Figure 4a and Table I, the CNN model trained without augmented data displayed over-fitting, yet performed better than the augmented data set. This is likely because the data set contains multiple slices of each patients' brain scan, and so the test, validation and training data sets may contain different slices from the same scans which have similar features. This means that over-fitting on the training data set translates to high accuracy on the test data set which would not be produced for a different data set. This could be improved by retrieving the patient IDs of each slice from the original data set on MATLab, so the test, validation and training data sets can be made to not contain any data from the same patients as each other.

The performance metrics displayed in Table I varied each time the code was ran due to the differences in the test, train and validation data sets. In order to get more accurate values k-fold cross-validation could be used, in which the dataset is split into k groups, and is trained k times, each time leaving a different one of these groups for validation, and using the rest for training. This may give results that better represent how the models would perform on real world data, however the similarity of images from multiple slices of each scan would need also need to be rectified to achieve this.[13]

Also, the accuracy could potentially be improved by experimenting more with the hyperparameters of Adam, such as weight decay. The need for a learning scheduler may also have been negated by enough optimisation of these hyperparameters, which would make the model simpler. Alternative optimisers could also be implemented, such as MADGRAD which consistently outperforms Adam and is not specially designed for any method or architecture, making it more versatile [14].

This model could also be extended to included a 'no tumour' category trained on healthy brain scans, making the model more versatile and useful for real world patient diagnosis.

Pretrained versions of ResNet-50 have been shown to be much more effective, producing AUCs of 1.00 across all categories, while keeping the computational cost reasonably low [1]. Further research could be extended to include these models to improve accuracy.

Also, attention mechanisms could be included to improve performance by making the model focus on the most important tumour characteristics for classification [15]. A Vision Transformer (ViT), which replaces the traditional CNN convolution layers with attention layers to improve accuracy with far less computational cost [15][16] could also be employed for this problem, as it has been shown to produce 99% accuracy and AUCs > 0.995 [17].

Use of Generative AI

Generative AI was used in this assignment for spotting errors in the code. For example, the network architectures were written by me and then copied into ChatGPT with the prompt 'are there any clear mistakes in these architectures', as an extra measure of troubleshooting if the code was not performing as expected. Mild improvements such as including a drop out rate were added as a result of generative AI suggestions. Also, the order of transformations was changed to resize the image first and turn it into a tensor last, as suggested by ChatGPT when I queried errors about the images not transforming to tensors, for example.

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